

H11 Division of labor vs herding is the sign of a Potts coupling, set by the goal

not yet rated

level: hypothesis · pipeline checks 76% · updated —

exploratory round 1 done (2026-10-03). The sign-by-goal-mode prediction is not supported. $\beta J_{CW} > 0$ (herding) in 11 of the 14 tested weeks, whatever the mode, including all five weeks predicted to be AF. The exceptions are the three weeks where agents built...

WHAT HAPPENED

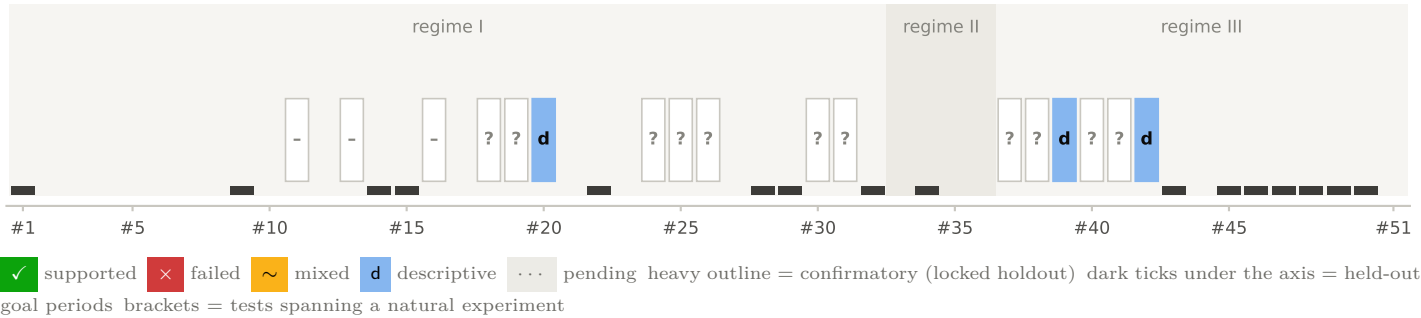
Not yet summarized. The question: When agents work on a shared goal, do they spread across subtasks (antiferromagnetic Potts coupling: division of labor, graph coloring) or pile onto the same project (ferromagnetic: herding)? Does consensus arrive as a first-order jump (mean-field Potts, $q \geq 3$)? Practical payoff: detecting duplicated work and group-think in multi-agent systems.

PHYSICAL MODEL

From: physics-models/10-potts, mean-field forward first. One uniform-coupling q-state Potts model per period, fitted from the dominant option’s share and its fluctuations. It predicts the sign of J by goal mode, and the jump size and hysteresis at consensus events. Full plmDCA couplings are only a secondary check (H02 showed pairwise inference is weak here). H11 variant (Curie–Weiss Potts with a neutral state, in roo

Pending.

GOAL PERIODS TESTED 3 descriptive, 3 n/a



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.